Question	Answer	Marks
6(a)	(a particle that) cannot be divided / subdivided (into smaller particles)	**B1**
6(b)(i)	any combination of one quark and one antiquark	**C1**
	up / charm / top **and** antiup / anticharm / antitop **or** down / strange / bottom **and** antidown / antistrange / antibottom	**A1**
6(b)(ii)	$E_{(K)} = \frac{1}{2}mv^2$	**C1**
	$2.1 \times 10^{-16} = \frac{1}{2} \times 0.67 \times 1.66 \times 10^{-27} \times v^2$	**C1**
	$v = 6.1 \times 10^5$ m s^{-1}	**A1**
6(c)(i)	number of electrons = 88	**A1**
6(c)(ii)	(number of nucleons = 228 – 5 × 4 = 208) number of protons = 88 – (5 × 2) + 4 = 82	**A1**
	number of neutrons= 208 – 82 = 126 = 126	**A1**